Rank-Aware Indigo-DPO: Scalable Preference Optimization for Industrial Talent Search Ranking

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Abstract

In global talent marketplaces, effective search ranking plays a pivotal role in driving hires and contributing millions in gross service volume (GSV). However, existing learning-to-rank (LTR) systems primarily optimize engagement metrics such as clicks, which fails to capture true hiring intent in scenarios where explicit conversions are extremely sparse ($<2.5\%$). Moreover, multilingual code-switching further introduces semantic ambiguity, leading to ranking instability and degraded hiring outcomes. To address these challenges, we propose **Rank-Aware Indigo-DPO**, a novel preference-optimized reranking framework tailored for multilingual talent search. The key idea is to integrate Direct Preference Optimization (DPO) with ranking-awareness to better align model predictions with downstream hiring objectives. Our approach introduces three innovations: (i) multi-negative preference sampling to stabilize optimization under sparse feedback, (ii) adaptive score calibration to mitigate multilingual noise, and (iii) a rank-aware margin loss that explicitly separates top candidates. Extensive experiments on real-world multilingual datasets demonstrate that Rank-Aware Indigo-DPO yields substantial improvements in both ranking precision and hiring coverage.

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Machine learning algorithms.*

Keywords

Talent Search, Preference Optimization, Reranking, LLMs, LoRA, Data Augmentation, Rank-aware

Figure 1: Traditional LTR vs. Indigo-DPO for "Web Scraping Data." LTR prioritizes high-CTR generalists, while Indigo-DPO ranks specialists with targeted expertise higher, aligning with hiring intent.

1 Introduction

Global talent marketplaces have become a cornerstone for connecting employers and job seekers, where effective search ranking plays a pivotal role in driving successful hires. In these systems, recruiters rely on ranking algorithms to retrieve the most suitable candidates from vast and heterogeneous pools. However, optimizing search quality is inherently challenging due to sparse conversion signals, multilingual query-candidate matching, and the complex interplay between user intent and observed engagement metrics.

Traditional learning-to-rank (LTR) models deployed in industrial talent search engines primarily optimize engagement metrics such as clicks, which often fail to capture true hiring intent when explicit conversions are extremely sparse ($<2.5\%$). Moreover, multilingual code-switching introduces semantic ambiguity, further degrading ranking stability and hiring outcomes. As illustrated in Fig. 1, when querying "*web scraping data*", a production LTR model incorrectly ranks a high-CTR generalist above a specialist with targeted expertise, due to click-optimization bias and multilingual noise. This mismatch highlights the core challenge we address: aligning ranking decisions with downstream hiring objectives rather than engagement proxies.

To tackle these challenges, we propose **Rank-Aware Indigo-DPO**, a scalable reranking framework extending Direct Preference Optimization (DPO) [30]. Indigo-DPO introduces three innovations: (i) *multi-negative sampling (S-DPO)*, enhancing training signals with multiple negatives for stable loss convergence; (ii) *adaptive score calibration (β-DPO)*, dynamically mitigating multilingual noise via entropy-based weighting; and (iii) *rank-aware margin loss*, ensuring confident separation of preferred candidates. The main contributions of this paper are summarized as follows:

- **Dynamic DPO Adaptation**: A novel integration of Direct Preference Optimization for freelance ranking, introducing β-calibrated loss to model real-time availability versus long-term quality trade-offs.
- **Efficient LLM Tuning**: A hybrid approach combining Low-Rank Adaptation (LoRA) [11] with softmax-based preference

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optimization, reducing training time and memory usage while preserving multilingual robustness.
- **Empirical and Business Impact**: Extensive experiments on large-scale multilingual datasets demonstrate that Rank-Aware Indigo-DPO consistently outperforms state-of-the-art LTR systems and LLM-based rerankers, yielding substantial improvements in both ranking precision and hiring coverage.

2 Preliminaries

2.1 Problem Statement

We study the second-stage reranking problem within industrial search systems. Given a textual query issued by a user, along with a set of initially retrieved candidates, our goal is to produce a refined ranking that closely matches the implicit preferences of the user. These implicit preferences are operationalized via explicit user actions, such as inviting or hiring a candidate. Formally, given a user query q and an initial candidate set $C = \{C_1, C_2, \ldots, C_k\}$ retrieved by a first-stage lightweight retrieval model, our reranking task aims to generate a refined ranked list, denoted as:

$$\text{rank}(q, C) \rightarrow \text{sorted candidate list.} \tag{1}$$

2.2 Notations and Definitions

Throughout this paper, we adopt the following notations to formally define the reranking task:
- **User Query (q)**: Textual input provided by the user to search candidates.
- **Job Context (d)**: Textual information describing the job posting, including its title and detailed description.
- **Interaction History (h)**: A sequence of historical interactions (e.g., clicks, invitations, hires) capturing implicit user preference signals.
- **Candidate Features (C_i)**: Each candidate's representation combining textual fields (e.g., profile title, summary) and structured metadata (e.g., Job Success Score (JSS), hourly rate, location, availability). Formally, we denote: $C_i = (c_i, \text{JSS}_i, \text{rate}_i, \ldots)$.

  For simplicity and clarity, we represent the full query context as:

$$Q = (q, d, h). \tag{2}$$

The reranking objective is to sort candidates C based on their relevance to Q, using a learned scoring function (see Section 3) defined over dense query–candidate embeddings.

3 Methodology

DPO formulates preference alignment as a contrastive learning problem: given a query and a pair of preferred and rejected outputs, the model is trained to assign higher likelihood to the preferred one. This approach eliminates the need for explicit reward modeling and supports stable, efficient fine-tuning using standard maximum likelihood setups. Moreover, DPO naturally extends to multi-negative settings and margin-aware formulations, making it a flexible foundation for industrial-scale reranking.

Our goal is to accurately predict user preferences in a talent search setting by modeling the rich contextual signals embedded in user queries, candidate profiles, and interaction histories. To this end, we propose **Indigo-DPO**, a preference-optimized reranking framework that builds on DPO and introduces three critical extensions tailored for noisy, multilingual, and sparse-feedback environments:
- **Multi-negative sampling** to exploit richer preference structures beyond pairwise contrast;
- **Adaptive calibration** to dynamically modulate the loss for low-confidence or noisy feedback;
- **Margin-aware regularization** to enforce confident separation between preferred and rejected candidates.

Figure 2 illustrates the overall framework, highlighting the interaction between preference data, contrastive loss, and the proposed enhancements.

3.1 Representation and Scoring

Given the context-aware query $Q = (q, d, h)$ and candidate set $C = \{C_1, C_2, \ldots, C_k\}$, our model first encodes these inputs into dense vector embeddings. Specifically, we use a dual-tower transformer-based architecture to obtain embeddings for both query context and candidate profiles.

Formally, the query context Q is encoded into a query embedding:

$$\mathbf{q} = f_\theta(Q) \in \mathbb{R}^d, \tag{3}$$

and each candidate C_i is encoded into a candidate embedding:

$$\mathbf{c}_i = g_\phi(C_i) \in \mathbb{R}^d, \quad i = 1, 2, \ldots, k, \tag{4}$$

where $f_\theta(\cdot)$ and $g_\phi(\cdot)$ represent embedding functions parameterized by transformer-based models with parameters θ and ϕ, respectively.

The relevance score between query context Q and candidate C_i is subsequently calculated using a dot-product scoring function:

$$s_i = \mathbf{q}^\top \mathbf{c}_i, \quad i = 1, 2, \ldots, k. \tag{5}$$

To capture normalized probabilities reflecting candidate selection likelihood, we apply a softmax function over these scores:

$$\pi_\theta(C_i \mid Q) = \frac{\exp(s_i)}{\sum_{j=1}^{k} \exp(s_j)}, \quad i = 1, 2, \ldots, k. \tag{6}$$

Ultimately, the reranking task is defined as generating a sorted list of candidates by their relevance scores:

$$\text{rank}(q, C) =_{C_i \in C} (s_i). \tag{7}$$

Prompt Construction. To effectively guide the LLM-based reranking model, we design structured textual prompts containing all relevant query and candidate information. An example of this prompt format is presented in Table 1. The prompt explicitly encodes multiple aspects crucial for accurate candidate ranking, including user queries (`QUERY`), associated job postings (`JOB_POST`), historical interactions (`HISTORY`), and detailed candidate attributes (`CANDIDATES`). During inference, the model consumes these structured prompts to predict the single best candidate (`OUTPUT`), directly aligning model outputs with observed user preferences.

3.2 Multi-Negative Preference Optimization

Effectively capturing user preferences requires careful handling of implicit preference signals. Conventional pairwise methods typically compare only one positive candidate against a single negative example, potentially limiting robustness and signal richness. To overcome this limitation, we propose an adaptive multi-negative

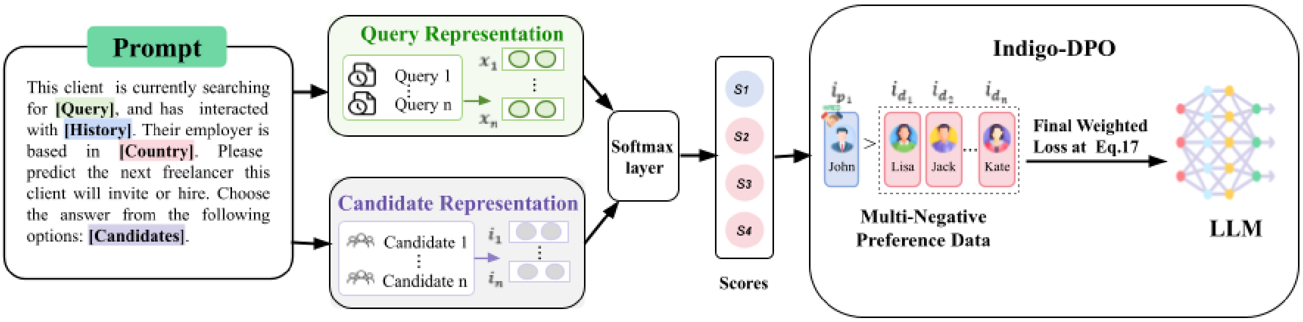

Figure 2: Overview of the Indigo-DPO framework. Indigo-DPO employs a dual-tower transformer architecture with softmax-based Direct Preference Optimization (DPO), incorporating multi-negative sampling (S-DPO), adaptive score calibration (β-DPO), and rank-aware margin loss to enhance candidate differentiation and ranking quality in real-world industrial talent search.

sampling strategy, wherein each training instance consists of a preferred candidate C^+ explicitly chosen by the user (e.g., invited or hired) and multiple negative candidates $\{C_1^-, C_2^-, \ldots, C_m^-\}$ concurrently displayed but not selected.

Formally, the multi-negative optimization loss is given by:

$$\mathcal{L}_{\text{S-DPO}} = -\frac{1}{m}\sum_{j=1}^{m} \log \frac{\exp(s_{C^+})}{\exp(s_{C^+}) + \exp(s_{C_j^-})}, \tag{8}$$

where s_{C^+} and $s_{C_j^-}$ represent the relevance scores calculated according to Eq. (5).

Recognizing the variability in data quality and signal strength across instances, we further introduce adaptive calibration weights β. These weights dynamically adjust the contribution of each training instance, helping to reduce the adverse effects of noisy, sparse, or multilingual preference signals. Specifically, we define:

$$\beta = \sigma\left(h_\psi(Q, C^+, \{C_j^-\})\right), \tag{9}$$

where $h_\psi(\cdot)$ is a small neural network parameterized by ψ, and $\sigma(\cdot)$ is a sigmoid function ensuring the calibration weight remains between 0 and 1. The adaptive loss then becomes:

$$\mathcal{L}_{\beta\text{-S-DPO}} = \beta \cdot \mathcal{L}_{\text{S-DPO}}. \tag{10}$$

This adaptive mechanism ensures higher robustness and improved model convergence by selectively emphasizing training instances with reliable signals. The β weights in β-DPO are learned from language entropy, prioritizing reliable signals in code-switched queries. This calibration enhances robustness across 30+ languages, addressing multilingual noise prevalent in global talent search

3.3 Margin-aware Regularization and Objective

To ensure clear and stable separation between positively and negatively labeled candidates, we incorporate a margin-aware hinge loss into our optimization framework. This margin constraint explicitly enforces a minimum difference δ between the scores of the positively selected candidate C^+ and the negatively selected candidates $\{C_j^-\}$, promoting better differentiation and robustness, especially among top-ranked candidates.

Formally, the margin-aware hinge loss is defined as follows:

$$\mathcal{L}_{\text{margin}} = \frac{1}{m}\sum_{j=1}^{m} \max\left(0, \delta - s_{C^+} + s_{C_j^-}\right), \tag{11}$$

where the hyperparameter δ explicitly specifies the desired minimum score separation.

The introduction of margin-aware regularization addresses the common issue of small margin differences observed in traditional softmax-based ranking methods. This regularization significantly enhances the discriminative power of the model, ensuring that genuinely preferred candidates consistently achieve noticeably higher scores compared to their negative counterparts.

Integrating both the adaptive multi-negative sampling loss and the margin-aware regularization, we define our comprehensive optimization objective as:

$$\mathcal{L}_{\text{Indigo-DPO}} = \mathcal{L}_{\beta\text{-S-DPO}} + \lambda \cdot \mathcal{L}_{\text{margin}}, \tag{12}$$

where λ is a hyperparameter controlling the relative importance of margin-aware regularization. Optimizing this objective ensures the model simultaneously maximizes preference alignment, robustness to noise, and meaningful candidate differentiation, ultimately yielding superior reranking performance in real-world talent search scenarios.

Table 1: Example serialized input prompt for Rank-Aware Indigo-DPO.

Field	Content
SYSTEM	You are a professional freelancer recommendation agent. Select exactly ONE freelancer matching client needs.
INSTRUCTIONS	(1) Read <QUERY>, <JOB_POST>, <HISTORY>, and <CANDIDATES>. (2) Compare candidates based on skills, JSS, hourly rate, and relevance. (3) Output single best match.
QUERY	`python web scraping`
JOB_POST	`PostID:88 | Title: Scrape real estate data weekly | Description: "Extract listing details weekly."`
HISTORY	`Last search "data scraper": viewed [101(JSS:0.96,Scrapy),102(JSS:0.89,Bs4)]; hired 101. Last invite: 103(JSS:0.92,Selenium).`

CANDIDATES	`1. ID:201 | Web Crawler Dev | JSS:0.95 | $10/hr | [Python,Scrapy]`
	`2. ID:202 | Python Automation Expert | JSS:0.91 | $12/hr | [Requests,Bs4]`
	`3. ID:203 | Full-stack Engineer | JSS:0.97 | $15/hr | [Node.js,React]`

OUTPUT	`Contractor ID: [NUMBER]`

4 Experiments and Results

We conducted extensive offline experiments using a proprietary dataset collected from real world freelancer platform, aimed at thoroughly validating the effectiveness of Rank-Aware Indigo-DPO. Specifically, our experiments are designed to answer the following research questions:

- **RQ1**: How does our proposed Rank-Aware Indigo-DPO perform compared to existing state-of-the-art reranking methods in terms of standard ranking metrics?
- **RQ2**: How effective is the rank-aware margin loss in improving the confidence margin between positive and negative candidates?
- **RQ3**: What are the incremental contributions of each introduced component to overall ranking performance?
- **RQ4**: Is Rank-Aware Indigo-DPO sufficiently efficient for practical, real-time deployment on an industrial-scale search platform?

4.1 Experimental Setup

Dataset. We constructed our evaluation dataset from 89,000 real-world search sessions conducted on real world freelancer platform during Jan–Mar 2025. Each session involved client-issued queries accompanied by at least one freelancer interaction event, such as an invitation or a confirmed hire. After rigorous data anonymization and quality filtering, our final dataset comprised 62,241 training samples, 13,337 validation samples, and 13,338 test samples. Approximately 42% of queries were in languages other than English, covering over 30 distinct languages, reflecting realistic, multilingual platform usage.

Training procedure. Rank-Aware Indigo-DPO is trained in two stages. We begin with an initial supervised fine-tuning (SFT) phase that leverages click and invite actions as soft engagement signals. Following this, we perform preference-based tuning using multi-negative sampling (S-DPO), adaptive β-calibration for noise reduction, and the rank-aware margin loss. We use LoRA-adapted LLaMA-2-7B (approximately 20M trainable parameters) trained on 8×A100 GPUs, with each epoch lasting about 18 hours.

Evaluation metrics. We measure model performance with three widely-used information retrieval metrics: Precision@1 (P@1, the fraction of queries whose top-ranked candidate matches the true preference), Normalized Discounted Cumulative Gain at position 5 (NDCG@5, measuring ranking quality in the top-5 results), and Mean Reciprocal Rank at position 5 (MRR@5, evaluating the average inverse rank of the first correct result). Statistical significance was determined via bootstrap resampling (1,000 samples, significance level $p < 0.05$).

4.2 Baseline Models

To comprehensively assess the improvements offered by Rank-Aware Indigo-DPO, we compared it against a series of representative baseline models:

- **BM25** [31]: A well-established lexical retrieval algorithm commonly employed in industry for its computational simplicity and low inference latency. BM25 relies entirely on keyword matching and ignores semantic or contextual signals.
- **RankNet** [5]: A classical pairwise learning-to-rank model based on neural networks, trained using logistic pairwise loss. RankNet efficiently models feature interactions but often struggles in scenarios with sparse or noisy user interactions.
- **ColBERT** [20]: A state-of-the-art neural retrieval model leveraging BERT-based embeddings for dense query-document representation. ColBERT provides robust semantic matching and strong accuracy, although it incurs high computational overhead, reducing its feasibility for real-time industrial deployment.
- **MonoT5** [27]: Employing the T5 transformer architecture, MonoT5 reranks candidates using powerful cross-attention mechanisms between queries and documents. MonoT5 typically delivers superior accuracy but at significant inference latency cost, limiting practical deployment scenarios.
- **Base LLM** [12, 34]: Our baseline uses the LoRA-adapted LLaMA-2 model without explicit preference-aware training. This baseline establishes a strong foundation for evaluating the incremental gains provided by preference optimization techniques.
- **Base LLM + DPO** [30]: This model extends the base LLM by explicitly optimizing predictions according to direct user preferences using standard DPO training. It significantly improves reranking accuracy and serves as our strongest baseline prior to incorporating rank-aware margin regularization.

4.3 Quantitative Results

We first address **RQ1** by systematically evaluating and comparing the performance of Rank-Aware Indigo-DPO against baseline methods. Results are summarized in Table 2.

Table 2: Ranking Results on Upwork Test Set

Model	P@1	NDCG@5	MRR@5
BM25	0.0572	0.3562	0.1867
RankNet	0.0867	0.3894	0.2234
ColBERT	0.1387	0.4943	0.3416
MonoT5	0.1321	0.4826	0.3312
Basic LLM	0.1450	0.5064	0.3520
Base DPO	0.1502	0.5221	0.3711
Indigo-DPO	**0.1710**	**0.5483**	**0.4223**

From the results, we observe that Indigo-DPO consistently outperforms all baseline methods across all metrics, demonstrating statistically significant improvements (all with $p < 0.05$). Specifically, Indigo-DPO notably improves P@1 by approximately 13.85%, NDCG@5 by 4.94%, and MRR@5 by 13.80% compared to the strongest baseline (Base DPO). These improvements confirm the effectiveness of explicitly modeling user preferences through adaptive multi-negative sampling and margin-aware regularization. Importantly, the substantial performance gain in P@1 (nearly 14%) indicates that Indigo-DPO particularly excels in scenarios where identifying the single best candidate is crucial, highlighting its utility in high-stakes hiring decisions.

To better understand the impact of our rank-aware margin loss, we examined the margin distributions defined as $\Delta s = s(c^+) - s(c^-)$. Figure 3 shows a substantial reduction in margin violations—instances where the margin $\Delta s < 1$—from 18.4% (Base DPO) to 11.7% (Rank-Aware Indigo-DPO). This reduction represents a relative drop of approximately **36.4%**, indicating stronger confidence and separation between positive and negative candidates.

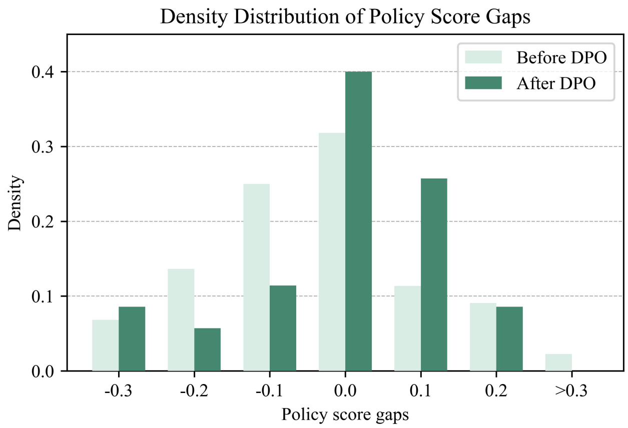

Figure 3: Margin distributions before vs. after applying rank-aware margin loss. Margin violations ($\Delta s < 1$) decrease significantly, reflecting improved ranking confidence.

4.4 Ablation and Component Analysis

We conducted a systematic ablation study to evaluate the contributions of each component introduced in Rank-Aware Indigo-DPO. As shown in Table 3, we observe consistent improvements at each stage:

- **S-DPO (multi-negative sampling)** stabilizes training by incorporating multiple negative examples, leading to modest gains (P@1 from 0.1502 to 0.1516).
- **β-DPO (adaptive calibration)** improves performance by reweighting noisy or multilingual samples.
- **Margin loss** introduces explicit preference separation and offers substantial gains.
- **Rank-aware margin loss** delivers substantial final gains, improving over Base DPO by 13.85% (P@1), 4.94% (NDCG@5), and 13.80% (MRR@5).

Table 3: Incremental contributions of Rank-Aware Indigo-DPO components (validation set).

Model Variant	P@1	NDCG@5	MRR@5
Base DPO	0.1502	0.5221	0.3711
+ S-DPO	0.1516	0.5270	0.3785
+ β-DPO	0.1528	0.5320	0.3842
+ Margin loss	0.1540	0.5379	0.3923
+ Rank-aware margin loss	**0.1710**	**0.5483**	**0.4223**

4.5 Efficiency and Practical Deployment

Finally, our method demonstrates promising practicality and efficiency based on extensive offline inference benchmarking and simulated deployment scenarios. Rank-Aware Indigo-DPO completes a full epoch of preference optimization training in approximately 18 hours using 8×A100 GPUs, making weekly retraining cycles feasible. Offline inference benchmarks—conducted under realistic production-like settings using T4 GPUs—indicate a median latency of approximately 40 ms and a 99th percentile latency of around 58 ms. Although these performance measures are currently based solely on offline simulation and inference tests, the results strongly suggest that Rank-Aware Indigo-DPO can comfortably meet the stringent latency requirements typical of industrial talent-search platforms.

5 Related Work

5.1 Learning-to-Rank models

Learning-to-Rank (LTR) models are widely used in web search [26, 32], e-commerce [2, 18], and recommendation systems [28, 38]. Classical approaches typically optimize surrogate metrics such as click-through rate (CTR) or NDCG through pointwise, pairwise [6], or listwise [7, 36] loss functions. However, these engagement-based metrics often misalign with critical downstream outcomes like hires and conversions [1, 17]. Recent frameworks, such as multi-objective learning (e.g., MMoE [25, 39]) and multitask approaches [22, 33], attempt to bridge this gap by optimizing multiple related objectives simultaneously. Additionally, rank-aware methods such as RankNet and LambdaMART [4, 16, 35] enhance ranking differentiation through margin-based losses. Nonetheless, these methods frequently overlook challenges related to explicit margin calibration, multi-negative handling, and multilingual adaptation, all of which are central to robust industrial deployments. In contrast, our Indigo-DPO method directly addresses these limitations by incorporating multi-negative sampling, adaptive calibration, and efficient parameter tuning, ensuring enhanced ranking confidence and practical applicability in dynamic, multilingual talent search scenarios.

5.2 LLM-Based Preference Reranking

Large Language Models (LLMs) have recently gained significant attention due to their capability to capture nuanced user preferences effectively at scale [9, 13]. Recent work has extended this capability to ranking and recommendation tasks using preference-aligned objectives such as Softmax-DPO [8], which improves model robustness via multi-negative contrastive learning. However, deploying these models in real-world industrial environments entails critical challenges, such as inference latency, training costs, and multilingual robustness [14, 15, 19]. Researchers have explored additional parameter-efficient techniques, including adapters, prefix-tuning, and prompt-tuning, to significantly reduce computational overhead without sacrificing performance [10, 12, 23, 29]. Recent work further extends LoRA to sequential recommendation by customizing LLMs instance-wise [21]. Furthermore, multilingual retrieval presents unique challenges. Litschko et al. [24] identify key difficulties and propose solutions for adapting neural ranking models to multilingual and dynamic user contexts—issues particularly characteristic of industrial talent marketplaces. To address these multilingual challenges, recent studies advocate strategic data augmentation and multilingual training strategies to enhance generalization across diverse languages and domains [3, 37].

6 Conclusion and Future Work

We presented **Rank-Aware Indigo-DPO**, a scalable preference-optimized reranking framework deployed in real world talent search system. By aligning results with observed conversion behavior (e.g., hires), the model leverages multi-negative sampling (S-DPO), adaptive calibration (β-DPO), and rank-aware margin loss to address label sparsity, multilingual variance, and structured-text fusion in real-world search environments. Fine-tuned via 4-bit LoRA, `Indigo-DPO` supports sub-60ms inference and ready to integrates into production pipeline. Evaluated on 62K real-world preference triples, `Indigo-DPO` achieves strong performance: P@1 = 0.1710 , NDCG@5 = 0.5483, and MRR@5 = 0.4223 over a base DPO model, outperforming MonoT5 and ColBERT. In offline tests on 5M sessions, pseudo-label augmentation improves hire conversion rates by 5.2% and reduces average hiring cycle time by 12%. We plan to extend Indigo-DPO to session-level modeling for multi-turn interactions, explore reward-aligned tuning via client feedback, and develop language-agnostic variants for code-switched multilingual search.

References

[1] Aman Agarwal, Ivan Zaitsev, Xuanhui Wang, Cheng Li, and Marc Najork. 2019. Estimating Position Bias Without Intrusive Interventions. In *Proceedings of the 12th ACM International Conference on Web Search and Data Mining (WSDM)* (Melbourne, Australia). ACM, 474–482. doi:10.1145/3289600.3291017

[2] Qingyao Ai, Keping Bi, Jiafeng Guo, and W. Bruce Croft. 2019. Learning a Neural Ranking Model with Weak Supervision. In *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 354–363. doi:10.1145/3331184.3331209

[3] Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2022. Efficient Cross-Lingual Transfer of Monolingual Language Models. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 320–332. doi:10.18653/v1/2022.acl-long.25

[4] Chris Burges. 2010. *From RankNet to LambdaRank to LambdaMART: An overview*. Technical Report. Microsoft Research.

[5] Christopher JC Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. 2005. Learning to Rank using Gradient Descent. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*. ACM, 89–96. doi:10.1145/1102351.1102363

[6] Christopher J. C. Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Gregory N. Hullender. 2005. Learning to Rank using Gradient Descent. *Proceedings of the 22nd International Conference on Machine Learning (ICML)* (August 2005), 89–96. doi:10.1145/1102351.1102363

[7] Zhe Cao, Tao Qin, Tie-Yan Liu, Ming-Feng Tsai, and Hang Li. 2007. Learning to Rank: From Pairwise Approach to Listwise Approach. In *Proceedings of the 24th International Conference on Machine Learning (ICML)*. ACM, 129–136. doi:10.1145/1273496.1273513

[8] Yuxin Chen, Junfei Tan, An Zhang, Zhengyi Yang, Leheng Sheng, Enzhi Zhang, Xiang Wang, and Tat-Seng Chua. 2024. On Softmax Direct Preference Optimization for Recommendation. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2406.09215

[9] Shijie Geng, Shuchang Liu, Zuohui Fu, Yingqiang Ge, and Yongfeng Zhang. 2022. Recommendation as Language Processing (P5): A Unified Pretrain, Personalized Prompt Predict Paradigm. In *Proceedings of the 16th ACM Conference on Recommender Systems*. ACM, 396–406. doi:10.1145/3523227.3546764

[10] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. 2019. Parameter-Efficient Transfer Learning for NLP. In *Proceedings of the 36th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 97)*. PMLR, 2790–2799. https://proceedings.mlr.press/v97/houlsby19a.html

[11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *International Conference on Learning Representations (ICLR)* (2021). https://arxiv.org/abs/2106.09685 Presented at ICLR 2022.

[12] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Weizhu Wang, and Yonghui Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *Proceedings of the Tenth International Conference on Learning Representations (ICLR)*. https://openreview.net/forum?id=rVLMp1wrnq

[13] Jianchao Ji, Zelong Li, Shuyuan Xu, Wenyue Hua, Yingqiang Ge, Juntao Tan, and Yongfeng Zhang. 2023. GenRec: Large Language Model for Generative Recommendation. *arXiv preprint arXiv:2307.00457* (2023). https://arxiv.org/abs/2307.00457

[14] Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models. *arXiv preprint arXiv:2310.05736* (2023). https://arxiv.org/abs/2310.05736

[15] Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2023. LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression. *arXiv preprint arXiv:2310.06839* (2023). https://arxiv.org/abs/2310.06839

[16] Thorsten Joachims. 2002. Optimizing Search Engines Using Clickthrough Data. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, 133–142. doi:10.1145/775047.775067

[17] Thorsten Joachims, Adith Swaminathan, and Tobias Schnabel. 2017. Unbiased Learning-to-Rank with Biased Feedback. In *Proceedings of the 10th ACM International Conference on Web Search and Data Mining (WSDM)* (Cambridge, UK). 781–789. doi:10.1145/3018661.3018699

[18] Md. Ahsanul Kabir, Mohammad Al Hasan, Aritra Mandal, Daniel Tunkelang, and Zhe Wu. 2024. A Survey on E-Commerce Learning to Rank. *arXiv preprint arXiv:2412.03581* (2024). https://arxiv.org/abs/2412.03581

[19] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling Laws for Neural Language Models. *arXiv preprint arXiv:2001.08361* (2020). https://arxiv.org/abs/2001.08361

[20] Omar Khattab and Matei Zaharia. 2020. ColBERT: Efficient and Effective Passage Search via Contextualized Late Interaction over BERT. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR)*. ACM, 39–48. doi:10.1145/3397271.3401075

[21] Xiaoyu Kong, Jiancan Wu, An Zhang, Leheng Sheng, Hui Lin, Xiang Wang, and Xiangnan He. 2024. Customizing Language Models with Instance-wise LoRA for Sequential Recommendation. In *Advances in Neural Information Processing Systems (NeurIPS)*. https://arxiv.org/abs/2408.10159

[22] Danwei Li, Zhengyu Zhang, Siyang Yuan, Mingze Gao, Weilin Zhang, Chaofei Yang, Xi Liu, and Jiyan Yang. 2023. AdaTT: Adaptive Task-to-Task Fusion Network for Multitask Learning in Recommendations. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. doi:10.1145/3580305.3599500

[23] Xiang Lisa Li and Percy Liang. 2021. Prefix-Tuning: Optimizing Continuous Prompts for Generation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*. Association for Computational Linguistics, 4582–4597. doi:10.18653/v1/2021.acl-long.353

[24] Robert Litschko, Ivan Vulić, and Goran Glavaš. 2022. Parameter-Efficient Neural Reranking for Cross-Lingual and Multilingual Retrieval. *arXiv preprint arXiv:2204.02292* (2022). https://arxiv.org/abs/2204.02292

[25] Jiaqi Ma, Zhe Zhao, Xinyang Yi, Jilin Chen, Lichan Hong, and Ed H. Chi. 2018. Modeling Task Relationships in Multi-task Learning with Multi-gate Mixture-of-Experts. In *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD)*. ACM, 1930–1939. doi:10.1145/3219819.3220007

[26] Bhaskar Mitra and Nick Craswell. 2018. An Introduction to Neural Information Retrieval. *Foundations and Trends® in Information Retrieval* 13, 1 (2018), 1–126. doi:10.1561/1500000061

[27] Rodrigo Nogueira and Jimmy Lin. 2020. Document ranking with a pretrained sequence-to-sequence model. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 708–718. doi:10.18653/v1/2020.emnlp-main.51

[28] Liang Pang, Jun Xu, Qingyao Ai, Xueqi Cheng, and Jirong Wen. 2020. SetRank: Learning a Permutation-Invariant Ranking Model for Information Retrieval. In *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*. 499–508. doi:10.1145/3397271.3401219

[29] Yujia Qin, Xiaozhi Wang, Yusheng Su, Yankai Lin, Ning Ding, Jing Yi, Weize Chen, Zhiyuan Liu, Juanzi Li, Lei Hou, Peng Li, Maosong Sun, and Jie Zhou. 2024. Exploring Universal Intrinsic Task Subspace for Few-Shot Learning via Prompt Tuning. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 32 (2024), 3631–3643. doi:10.1109/TASLP.2024.3430545

[30] Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct Preference Optimization: Your Language Model is Secretly a Reward Model. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. https://arxiv.org/abs/2305.18290

[31] Stephen Robertson and Hugo Zaragoza. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval* 3, 4 (2009), 333–389. doi:10.1561/1500000019

[32] Stephen Robertson and Hugo Zaragoza. 2009. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends in Information Retrieval* 3, 4 (2009), 333–389. doi:10.1561/1500000019

[33] Jianing Sun, Yingxue Zhang, Chen Ma, Mark Coates, Huifeng Guo, Ruiming Tang, and Xiuqiang He. 2019. Multi-Graph Convolution Collaborative Filtering. In *2019 IEEE International Conference on Data Mining (ICDM)*. IEEE, 1306–1311. doi:10.1109/ICDM.2019.00170

[34] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. 2023. Llama 2: Open Foundation and Fine-Tuned Chat Models. *arXiv preprint arXiv:2307.09288* (2023). https://arxiv.org/abs/2307.09288

[35] Jason Weston, Samy Bengio, and Nicolas Usunier. 2011. WSABIE: Scaling up to large vocabulary image annotation. In *Proceedings of the Twenty-Second International Joint Conference on Artificial Intelligence (IJCAI)*. AAAI Press, 2764–2770.

https://www.ijcai.org/Proceedings/2011/384
[36] Fen Xia, Tie-Yan Liu, Jue Wang, Wensheng Zhang, and Hang Li. 2008. Listwise Approach to Learning to Rank: Theory and Algorithm. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*. ACM, 1192–1199. doi:10.1145/1390156.1390306
[37] Linting Xue. 2022. Multilingual Adaptation for Large Language Models. (2022).
[38] Shuai Zhang, Lina Yao, Aixin Sun, and Yi Tay. 2019. Deep Learning based Recommender System: A Survey and New Perspectives. *ACM Computing Surveys (CSUR)* 52, 1 (2019), 1–38. doi:10.1145/3285029
[39] Zhe Zhao, Lichan Hong, Li Wei, Jilin Chen, Aniruddh Nath, Shawn Andrews, Aditee Kumthekar, Maheswaran Sathiamoorthy, Xinyang Yi, and Ed H. Chi. 2019. Recommending What Video to Watch Next: A Multitask Ranking System. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*. ACM, 43–51. doi:10.1145/3298689.3346997

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